

ZHAOZHOU LI

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EMPLOYMENT

• Nanjing University, China	Associate Professor	2025 –
• Hebrew University of Jerusalem, Israel	Marie Skłodowska-Curie Fellow	2023 – 2025
• Hebrew University of Jerusalem, Israel	Postdoctoral Fellow	2021 – 2023
• Shanghai Jiao Tong University, China	Postdoctoral Researcher	2018 – 2021

EDUCATION

• Shanghai Astronomical Observatory, China	Ph.D. in Astrophysics	2011 – 2017
• Beihang University, China	B.S. in Applied Physics	2007 – 2011

RESEARCH TOPICS

Dynamics and formation of cosmic structures from the Early Universe to the Local Group

- Star and galaxy formation at cosmic dawn: feedback-free starbursts
- Galaxy structural evolution and diversity
 - Dark matter (DM) halos: dynamical structure and interplay with baryons and feedback
 - Satellite galaxies: initial orbits and mass function, tidal evolution, and fate
 - Formation of DM-deficient galaxies and ultra-diffuse/compact galaxies
- Novel dynamical modeling methods: mass and boundary of the Milky Way (MW) halo
- Novel robust Gaussian process, with application to modeling binary stars in star clusters

PROFESSIONAL EXPERTISE

- Cosmological/isolated simulations & analysis
Merger tree, (sub)structure finding, large-scale structure, GADGET, semi-analytical models
- Observational data analysis
Analysis of survey catalogs (SDSS/BOSS, *Gaia*), modeling resolved stellar populations
- Statistics and machine learning
Advanced Bayes, Gaussian process, robust statistics, clustering, active learning, etc
- Programming (Expert – Python; familiar – C, Fortran, SQL)
High performance computing (parallel Python, C, Cython), numerical analysis (Scipy, GSL), big data reduction

HONORS AND AWARDS

• Marie Skłodowska-Curie Actions Fellowship, €200,000, Europe Commission	2023 – 2025
• Rosenblum Award for Excellence in Astrophysics, \$2,500 (travel fund), HUJI	2022
• Second Prize of the National Mathematics Competitions for College Students, China	2010
• First Prize of the Physics Experiment Competition for College Students, Beijing	2009
• Outstanding Freshman Scholarship, Second Prize, Beihang University	2007

PROPOSALS AND GRANTS

- Co-Investigator on NSF-BSF Research Grant (2307290; PI: van den Bosch; Dekel) 2024 – 2027
Getting to the Core of Core Formation & Core Dynamics, \$238,000 (BSF) + \$490,000 (NSF)
Key contributor to the theoretical basis of the proposal
- Co-Investigator on NSF-BSF Research Grant (2406558; PI: Teyssier; Dekel) 2024 – 2027
The Origin of the Excess of Bright Galaxies at Cosmic Dawn, \$233,000 (BSF) + \$539,000 (NSF)
Responsible for analysis involving semi-analytical models
- Marie Skłodowska-Curie Actions Fellowship, European Commission (101109759) 2023 – 2025
Dark Matter Cusps and Cores by Violent Relaxation, €200,000
Independent postdoc fellow; **proposal scored 99/100, top 1% of 800** proposals in physics
- Participant in Chinese Space Station Telescope Project (CMS-CSST-2021-B03) 2020 – 2022
The Composition & Structure of the Milky Way & Local Group with CSST, ¥3.96M (~\$546,700)
Support in dynamical modeling methods

TEACHING

- Guest lecturer Advanced Cosmology (graduate course, 4h cumulative) HUJI 2022S, 2023S
Introductory lecture on galactic dynamics
- Guest instructor Astrophysics Seminar (undergraduate training, 20h) HUJI 2022S
Research training: reading, presenting, and report writing on cosmological challenges
- Lecturer/advisor Scientific Practice Projects (high school students, 100h) 上海 Shanghai 2016 – 2018
Reproducing Hubble's Law: data analysis, presentation, and report writing
- Lecturer Intro to Astronomy (primary & middle schools, 25h) Shanghai 2015 – 2017
Night sky, Earth, Moon, and Planets
- Lecturer Workshop: Applied Python in Astronomy (4h) Shanghai Nov 2015
Computation, data reduction, and visualization

ADVISING (GRADUATES)

- Close advising: Qi Guo (Kavli IPMU, modeling filaments), Yarone Tokayer (Yale, halo dynamics)
- As scientific co-advisor: Yaofei He (THU, halo response to feedback), Feihong He (SJTU, subhalo evolution), Yanrui Zhou (SJTU, dynamical modeling), Axel Gross (UMN, halo structure)
- As technical co-advisor: Rui Shi (SJTU), Xiaokai Chen (SJTU), Xianguang Meng (SHAO)

ACADEMIC SERVICE

- Referee for scientific journals: MNRAS, ApJ, Universe Since 2022
- Coordinator of the astrophysics seminar at HUJI 2022 – 2025
- LOC member of the conference *Studying the Universe with Galaxy Surveys*, Shanghai 2018
- Founder and maintainer of the AstroPython wechat discussion groups (~1000 users) Since 2016
- Maintainer of the computing servers of the cosmology group at SHAO 2014 – 2020
- Coordinator of the cosmology journal club at SHAO and SJTU 2014 – 2016, 2018 – 2019

OUTREACH

- Public lecture at a book club, “Ramble under the Starry Sky” (2h), Changsha Jul 2023
- Expositor of the Open Day of Physics and Astronomy (20h), SJTU 2017 – 2019
- Volunteer guide at the Shanghai Natural History Museum 2016
- Member of the Interplanetary Immigration Agency, a near-future science fiction project 星际 Since 2014
- Co-organizer of sidewalk astronomy nights and stargazing camps (> 20), Beijing 2007 – 2011

OPEN-SOURCE CONTRIBUTION

- Personal codes cited in ~70 papers across various disciplines (exoplanets, bioinformatics, agriculture, etc.) 🎓
- Selected personal software, see more at <https://syrte.github.io/code>
 - cyper: running Cython codes on the fly for high performance Python 🔗
 - robustgp: proposed novel Gaussian process regression for contaminated data (~20 citations) 🔗
 - ndtest: multi-dimensional statistical tests (incl. 2D K-S test, >50 citations) 🔗
 - ParsecQuery: query isochrones from the PARSEC stellar evolution model 🔗
 - virgo_query: query data from Virgo simulation database 🔗
- Contributor to infrastructure libraries including Numpy, Scipy, Cython
- Ranking by public contribution:
 - Top 10% overall on StackOverflow 📖, posts with ~2.9M people reached
 - Top 500 in Israel on GitHub 🔗

SOCIAL SERVICE

- Coordinator for *Voice of Flowing Hearts*, exhibition featuring photos by migrant children, Beijing 2010
- Volunteer for welfare field survey of schools for children of rural migrant workers, Beijing 2010
- Disaster volunteer of the Sichuan Earthquake (1 month), Pengzhou 2008

SEMINAR TALKS

- Weizmann Institute of Science, Israel Apr 2025
- SWIFAR, Yunan University, China Mar 2025
- Tsinghua University, China Jul 2024
- Shanghai Jiao Tong University, China Jun 2024
- Nanjing University, China Jun 2024
- Ben-Gurion University, Israel Jan 2024
- Purple Mountain Observatory, China Dec 2023
- Nanjing University, China Dec 2023
- Shanghai Normal University, China Dec 2023
- CCA, Flatiron Institute, US Oct 2023
- Yale University, US Oct 2023
- National Astronomical Observatories, China (×2) Jul 2023
- Shanghai Astronomical Observatory, China Jun 2023
- University of Minnesota, US Nov 2022
- Hebrew University of Jerusalem, Israel Apr 2022
- Hebrew University of Jerusalem, Israel Mar 2021
- Kavli IPMU, University of Tokyo, Japan Sep 2020
- KIAA, Peking University, China Jun 2020
- Shanghai Astronomical Observatory, China Apl 2020
- SWIFAR, Yunnan University, China Nov 2019
- ICC, Durham University, UK Jul 2019
- Kavli IPMU, University of Tokyo, Japan Aug 2018
- Shanghai Jiao Tong University, China Nov 2017

CONFERENCE PRESENTATIONS

- Santa Cruz Galaxy Workshop Aug 2025
- International Workshop on Galaxy Formation + AGORA in Asia 2025, Osaka (remote) May 2025
From FFB starbursts at cosmic dawn to massive quiescent galaxies at cosmic morning (presented on behalf of Prof. A. Dekel)
- Expanding the boundaries of dark matter halo, Shanghai (invited) May 2025
Dynamics of halos: Boundary, accretion and tidal stripping (remote talk due to visa issues)
- Galaxy origins in the JWST era: a “crisol” of stars, ISM, and SMBHs, Toledo May 2025
Formation of galaxies at cosmic dawn by feedback-free starbursts (remote talk due to visa issues)
- Three Years of JWST: New Observations, New Challenges, Nanjing (invited) Dec 2024
- Cosmic Dawn Revealed by JWST: First Stars, Galaxies, and Black Holes, Santa Barbara Aug 2024
Feedback-free starbursts at cosmic dawn: Observable predictions for JWST
- 32nd IAU General Assembly 2024, Cape Town Aug 2024
Analytical insights into Dark Matter Deficient Galaxies (remote talk due to visa issues)
- Santa Cruz Galaxy Workshop (invited) Jul 2024
Analytical insights into Feedback Free Starbursts
- Santa Cruz Galaxy Workshop (invited) Aug 2023
- Collaboration Workshop on Cosmology and Galaxy Formation, Shanghai (invited) Jun 2023
- DDA54: Annual Meeting of the Division on Dynamical Astronomy of AAS, Remote May 2023
Modeling the formation of dark-matter deficient galaxies
- AI for Astronomy, Shenzhen/Online Nov 2022
Robust Gaussian process and its application to resolved stellar population
- Santa Cruz Galaxy Workshop (invited) Aug 2022
Modeling the Response of Halos to Gas Ejection and Tidal Stripping
- DDA53: Annual Meeting of the Division on Dynamical Astronomy of AAS, Remote Apr 2022
Modeling the response of dark matter haloes to gas ejection
- EAS: European Astronomical Society Annual Meeting, Online Jul 2021
Measuring the Milky Way mass profile from satellite galaxies kinematics
- DDA52: Annual Meeting of the Division on Dynamical Astronomy of AAS, Online May 2021
A novel dynamical modeling method based on the data-driven distribution function
- Guoshoujing Meeting on Galaxies and Cosmology, Hangzhou May 2021
The outer edges of the Milky Way halo from the motion of nearby galaxies
- Cross-Strait Symposium on Star Cluster Studies, Online Dec 2020
Precise determination of the main sequence of open clusters in the CMD
- Chinese Astronomical Society Annual Meeting, Online Oct 2020
- Shanghai Assembly on Cosmology and Galaxy Formation, Shanghai Nov 2019
Constrain the Milky Way Mass Profile with Phase Space Distribution of Satellite Galaxies
- Galaxy Angular Momentum Alignment 2019, Shanghai Oct 2019
Satellite Kinematics and Milky Way Halo Mass
- The Milky Way 2019: LAMOST and Other Leading Surveys, Yichang Oct 2019
Measure the Milky Way Mass Profile with Satellite Galaxies in Phase Space
- Small Galaxies, Cosmic Questions, Durham (poster talk) Jul 2019
Milky Way Mass Profile from Satellite Dynamics
- Astrophysical Dynamics, Tsung-Dao Lee Institute, Shanghai Jul 2019
- Galactic Dynamics in the Era of Large Surveys, Shanghai Jul 2019
Measure the Milky Way Mass Profile with Satellite Galaxies in Phase Space
- Halo and Galaxy Assembly Bias — from Theory to Observation, Shanghai Jun 2019

- The Life and Times of the Milky Way, Shanghai Nov 2018
- Studying the Universe with Galaxy Surveys Revealing the Unlimited in Shanghai Jun 2018
Milky Way Halo Mass from Satellite Kinematics
- SHAO-PKU Bilateral Symposium, Shanghai Aug 2017
- 11th Zhang Heng Meeting of the Chinese Astronomical Society, Guiyang Jun 2017
Determination of Milky Way Halo Mass from Kinematics of Satellite Galaxies

PUBLICATIONS

Since 2017: 40 papers (10 first author, 10 second/third author), 2 proceedings
1000 citations, H=16 [ADS , arXiv , Google Scholar 

As first/second author with major contributions (12)

40. Scaling relations in the phase space structure of dark matter haloes
Gross, A., **Li, Z.**, and Qian, Y.-Z., 2025, MNRAS, 538, 1843 
39. emPDF: Inferring the Milky Way mass with data-driven distribution function in phase space
Li, Z., Han, J., Wang, W., Qian, Y.-Z., Li, Q., Jing, Y., and Li, T.S., 2025, MNRAS, 538, 1442 
38. Phase space distribution functions and energy distributions of dark matter particles in haloes
Gross, A., **Li, Z.**, and Qian, Y.-Z., 2024, MNRAS, 530, 836 
37. Feedback-free starbursts at cosmic dawn: Observable predictions for JWST
Li, Z., Dekel, A., Sarkar, K.C., Aung, H., Giavalisco, M., Mandelker, N., and Tacchella, S., 2024, A&A, 690, A108 
36. The response of dark matter haloes to gas ejection: CuspCore II
Li, Z., Dekel, A., Mandelker, N., Freundlich, J., François, T., 2023, MNRAS, 518, 5356 
35. The outermost edges of the Milky Way halo from galaxy kinematics
Li, Z. and Han, J., 2021, ApJL, 915, L18 
34. Robust Gaussian process regression based on iterative trimming
Li, Z., Li, L., and Shao, Z., 2021, Astronomy and Computing, 36, 100483  
33. Orbital distribution of infalling satellite halos across cosmic time
Li, Z., Zhao, D.-H., Jing, Y.P., Han, J., and Dong, F.-Y., 2020, ApJ, 905, 177 
32. Constraining the Milky Way mass profile with phase-space distribution of satellite galaxies
Li, Z., Qian, Y.-Z., Han, J., Li, T.S., Wang, W., and Jing, Y.P., 2020, ApJ, 894, 10 
31. A versatile and accurate method for halo mass determination from phase-space distribution of satellite galaxies
Li, Z., Qian, Y.-Z., Han, J., Wang, W., and Jing, Y.P., 2019, ApJ, 886, 69 
30. The structure finders and the subhalo population in cosmological simulations (*Review in Chinese*)
Li, Z., Han, J.-X., 2018, Progress in Astronomy, 36-3, 306  (w/ English abstract)
29. Determination of dark matter halo mass from dynamics of satellite galaxies
Li, Z., Jing, Y.P., Qian, Y.-Z., Yuan, Z., and Zhao, D.-H., 2017, ApJ, 850, 116 

As coauthor with significant contributions

28. From FFB Starbursts at Cosmic Dawn to Quenching at Cosmic Morning: Hi-z Galaxy Bimodality
Dekel, A., Mandelker, N., **Li, Z.**, Yao, Z., Zhu, B., Lapiner, S., Dutta Chowdhury, D., and Ginzburg, O., 2025, arXiv, arXiv:2506.11664 
27. Growth of massive black-holes in ffb galaxies at cosmic dawn
Dekel, A., Stone, N.C., Dutta Chowdhury, D., Gilbaum, S., **Li, Z.**, Mandelker, N., and van den Bosch, F.C., 2025, A&A, 695, A97 
26. Why artificial disruption is not a concern for current cosmological simulations
He, F., Han, J., and **Li, Z.**, 2025, ApJ, 981, 108 
25. How do the velocity anisotropies of halo stars, dark matter, and satellite galaxies depend on host halo

properties?

He, J., Wang, W., **Li, Z.**, Han, J., Rodriguez-Gomez, V., et al., 2024, ApJ, 976, 187 

24. Are Odd Radio Circles virial shocks around massive galaxies? Implications for cosmic-ray diffusion in the circumgalactic medium
Yamasaki, S., Sarkar, K.C., and **Li, Z.**, 2024, MNRAS, 528, 3854 
23. Efficient Formation of Massive Galaxies at Cosmic Dawn by Feedback-Free Starbursts
Dekel, A., Sarkar, K.S., Birnboim, Y., Mandelker, N., and **Li, Z.**, 2023, MNRAS, 523, 3201 
22. Is the core-cusp problem a matter of perspective: Jeans Anisotropic Modeling against numerical simulations
Wang, W., Zhu, L., **Li, Z.**, Chen, Y., Han, J., He, F., Yang, X., et al., 2022, ApJ, 941, 108 
21. The universal specific merger rate of dark matter halos
Dong, F., Zhao, D., Han, J., **Li, Z.**, Jing, Y., and Yang, X., 2022, ApJ, 929, 120 
20. A machine learning approach to infer the accreted stellar mass fractions of galaxies
Shi, R., Wang, W., **Li, Z.**, et al., 2022, MNRAS, 515, 3938S 
19. Modeling Unresolved Binaries of Open Clusters in the Color-Magnitude Diagram. I. Method and Application of NGC 3532
Li, L., Shao, Z., **Li, Z.**, Yu, J., Zhong, J., and Chen, L., 2020, ApJ, 901, 49 
18. The mass of our Milky Way (*Review*)
Wang, W., Han, J., Cautun, M., **Li, Z.**, and Ishigaki, M.N., 2020, Science China: Physics, Mechanics & Astronomy, 63, 109801 

Collaboration papers

17. Overmassive Black holes live in compact galaxies in the early Universe
Wu, Y., Wang, T., Liu, D., Tan, Q., Ho, L.C., Zhang, Z., Shi, Y., Xu, K., Kohno, K., Wang, R., Izumi, T., and **Li, Z.**, 2025, arXiv, arXiv:2506.14896 
16. Insights on Metal Enrichment and Environmental Effect at $z \approx 5 - 7$ with JWST ASPIRE/EIGER and Chemical Evolution Model
Li, Z., Kakiichi, K., Christensen, et al. (incl. **Li, Z.**), 2025, arXiv, arXiv:2504.18616 
15. Pushing JWST to the extremes: search and scrutiny of bright galaxy candidates at $z \simeq 15-30$
Castellano, M., Fontana, A., Merlin, E., et al. (incl. **Li, Z.**), 2025, arXiv, arXiv:2504.05893 
14. The rise of the galactic empire: luminosity functions at $z \sim 17$ and $z \sim 25$ estimated with the MIDIS+NGDEEP ultra-deep JWST/NIRCam dataset
Pérez-González, P.G., Östlin, G., Costantin, et al. (incl. **Li, Z.**), 2025, arXiv:2503.15594 
13. The mass and redshift dependence of halo star clustering
Tan, Z., Wang, W., He, J., Zhang, Y., Rodriguez-Gomez, V., Han, J., **Li, Z.**, and Yang, X., 2024, ApJ, 976, 19 
12. Inferring the mass content of galaxy clusters with satellite kinematics and Jeans Anisotropic modeling
Shi, R., Wang, W., **Li, Z.**, Zhu, L., Smith, A., Cole, S., Gao, H., et al., 2024, ApJ, 973, 82 
11. JWST/MIRI Reveals the True Number Density of Massive Galaxies in the Early Universe
Wang, T., Sun, H., Zhou, L., Xu, K., Cheng, C., **Li, Z.**, et al., 2025, ApJL, 988, L35 
10. Evidence for a shallow evolution in the volume densities of massive galaxies at $z=4$ to 8 from CEERS
Chworowsky, K., Finkelstein, S.L., Boylan-Kolchin, M., et al. (incl. **Li, Z.**), 2024, AJ, 168, 113 
9. Effects of feedback-free starburst galaxies on the 21-cm signal and reionization history
Libanore, S., Flitter, J., Kovetz, E.D., **Li, Z.**, and Dekel, A., 2024, MNRAS, 532, 149 
8. DESI Legacy Imaging Surveys Data Release 9: cosmological constraints from galaxy clustering and weak lensing using the minimal bias model
Xu, H., Li, H., Zhang, J., et al. (incl. **Li, Z.**), 2023, Science China: Physics, Mechanics & Astronomy, 66, 129811 
7. Unraveling the Complexity of Dwarf Galaxy Dynamics: A Study of Binary Orbital Motions
Wang, W., Zhu, L., Jing, Y., Grand, R.J.J., **Li, Z.**, et al., 2023, ApJ, 956, 91 
6. Physical evolution of dark matter halo around the depletion boundary

Gao, H., Han, J., Fong, M., Jing, Y.P., and **Li, Z.**, 2023, ApJ, 953, 37 

5. The growth pattern of liver metastases on MRI predicts early recurrence in patients with colorectal cancer: a multicenter study
Cai, Q., Mao, Y., Dai, S., et al. (incl. **Li, Z.**), 2022, European Radiology, 32, 7872 
4. What to expect from dynamical modelling of cluster haloes - I. The information content of different dynamical tracers
Li, Q., Han, J., Wang, W., Cui, W., **Li, Z.**, and Yang, X., 2021, MNRAS, 505, 3907 
3. Weak equivalence principle, swampland and H_0 tension with fast single radio bursts FRB 180924 and FRB 190523
Wang, D., **Li, Z.**, and Zhang, J., 2020, Physics of the Dark Universe, 29, 100571 
2. The first constraint from SDSS galaxy-galaxy weak lensing measurements on interacting dark energy models
Zhang, J., An, R., Luo, W., **Li, Z.**, Liao, S., and Wang, B., 2019, ApJL, 875, L11 
1. Fully self-consistent cosmological simulation pipeline for interacting dark energy models
Zhang, J., An, R., Liao, S., Luo, W., **Li, Z.**, and Wang, B., 2018, Phy. Rev. D, 98, 103530 

Conference proceedings

2. Dynamical interaction in the stellar cluster – Evidence from binaries of NGC3532
Li, L., Shao, Z., **Li, Z.**, 2021, Joint Statistical Meetings (JSM) proceedings, 2021.317202
1. Satellite galaxies as better tracers of the Milky Way halo mass
Han, J., Wang, W., and **Li, Z.**, 2020, Galactic Dynamics in the Era of Large Surveys, IAU Symposium, 353, 109 